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Machine learning post-processing for air quality prediction in Japan

Keiya Yumimoto^{1*} , Syuichi Itahashi¹, Masamitsu Hayasaki² and Seiji Sugata³

Abstract

A deep learning-based post-processing (PP) system was developed to improve the predictive skill for ozone and particulate matter less than 2.5 μm in diameter ($\text{PM}_{2.5}$) forecasts in the Visual atmospheric Environment Utility System (VENUS) operational air quality forecasting system in Japan. The PP system uses a deep neural network trained on paired Weather Research and Forecasting (WRF)/Community Multi-scale Air Quality Modeling System (CMAQ) model outputs and ground-based observations from 2013 to 2016, with its performance evaluated using independent data from 2017. For ozone, PP removed the systematic overestimation bias in the raw forecasts, reducing the mean deviation from 16.5 to -0.6 ppbv; increased the domain-wide correlation coefficient from 0.657 to 0.870; and decreased the root mean square deviation (RMSD) markedly from 20.7 to 8.1 ppbv. For $\text{PM}_{2.5}$, PP improved predictive skill, increasing the domain-wide correlation coefficient from 0.544 to 0.671 and reducing the RMSD from 9.0 to 6.5 $\mu\text{g}/\text{m}^3$. However, the long-term decrease in observed concentrations can cause overcorrection in the fixed PP model. The PP system corrected errors arising from uncertainties in physical and chemical parameterizations directly and limited model resolution at a lower computational cost than data assimilation. Our findings demonstrate that the PP system is suitable for operational implementation and that strategies such as sequential retraining are important for including long-term concentration trends.

Keywords DNN, Machine learning, Post-processing, Air quality forecasting, Ozone, $\text{PM}_{2.5}$

1 Introduction

Ambient air pollution, particularly fine particulate matter less than 2.5 μm in diameter ($\text{PM}_{2.5}$) and tropospheric ozone, elevates the risk of respiratory and cardiovascular diseases and can have substantial adverse effects on human health and socioeconomic activities (Cohan et al. 2017; World Health Organization 2024). To reduce these effects, it is essential to have accurate air quality forecasts for several days to about 1 week ahead as well as early

warning systems that make use of these forecasts (Kelly et al. 2012). Many countries have implemented operational air quality forecasting systems that couple numerical weather prediction models with chemical transport models (CTMs) (e.g., NOAA and U.S. EPA 2025; Copernicus Atmosphere Monitoring Service (CAMS) 2024; National Institute for Environmental Studies (NIES) 2025). However, because of uncertainties in emission inventories and meteorological fields and approximations in physical and chemical parameterizations, CTM outputs still have systematic forecast biases and limited skill in reproducing high-pollution episodes (Cha et al. 2025; Guérette et al. 2020; Li et al. 2019; Chen et al. 2019).

Data assimilation and emission inversion are powerful tools for addressing these problems and increasing the skill of air quality forecasts (Bocquet et al. 2015; Elbern et al. 2007; Yumimoto et al. 2018). However, the development and operation of chemical data assimilation

*Correspondence:

Keiya Yumimoto
yumimoto@iam.kyushu-u.ac.jp

¹ Research Institute for Applied Mechanics, Kyushu University, 6-1 Kasuga-Koen, Kasuga, Fukuoka 816-8580, Japan

² Japan Automobile Research Institute (JARI), 2530 Karima, Tsukuba, Ibaraki 305-0822, Japan

³ National Institute for Environmental Studies, 16-2 Onagawa, Tsukuba, Ibaraki 305-8506, Japan

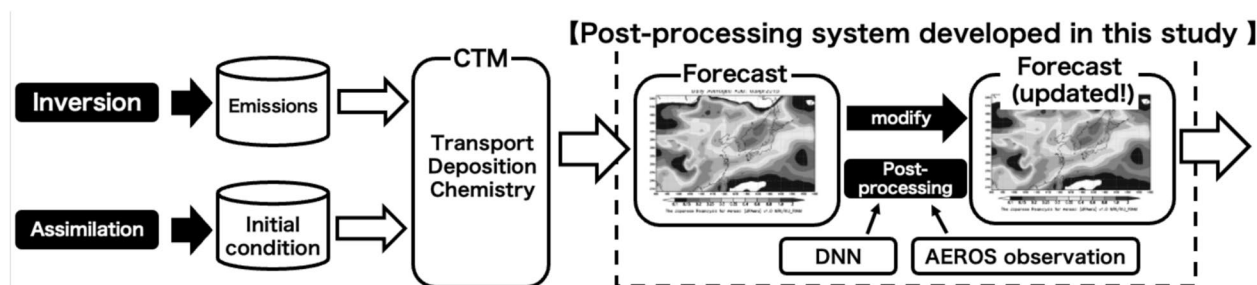


Fig. 1 An overview of the PP system developed in this study

systems are computationally demanding and resource intensive, and thus implementing these tools in real-time forecasting systems and maintaining them in routine operations remains challenging (Bocquet et al. 2015). Therefore, complementary approaches are required that can improve forecasts with a smaller burden on computational and human resources.

In numerical weather prediction, statistical post-processing (PP; also called statistical guidance) is widely used to correct systematic errors and spatial biases in model forecasts and has substantially improved forecasting accuracy (Wilks 2011; Gneiting et al. 2005; Vannitsem et al. 2021). PP constructs empirical correction relationships between past forecasts and observations and applies the relationships in a PP step to adjust the raw model output. In contrast, data assimilation updates the model initial conditions or emissions using observations. Because PP is computationally inexpensive and easy to add to an existing forecasting system, this approach has become a standard component of many operational numerical weather prediction suites. Motivated by these applications in numerical weather prediction, similar PP approaches based on machine learning and deep learning techniques that integrate CTM output with observational data have been proposed for air quality forecasting (Sayeed et al. 2021; Stergiou et al. 2025). Sayeed et al. (2021) developed a generalized deep convolutional neural network that uses Community Multi-scale Air Quality Modeling System (CMAQ) outputs, meteorological fields, and surface observations to bias-correct and extend $PM_{2.5}$, PM_{10} , and nitrogen dioxide forecasts from 1 to 7 days ahead over South Korea. They demonstrated that the convolutional neural network PP outperformed the raw CTM even at longer lead times. Stergiou et al. (2025) developed a hybrid deep learning model that combines convolutional neural networks and long short-term memory to post-process CMAQ ozone forecasts at 10 sites in Texas, improving bias, high-ozone event detection, and temporal pattern alignment compared with the raw model. Nevertheless, research on these approaches

is still at an early stage. Studies have mainly been site-specific and localized to particular regions or monitoring stations. Furthermore, methodological heterogeneity persists, including non-standardized selections for the target species, forecast lead times, design of input variables, and evaluation metrics. Consequently, the degree of generalizability achievable by machine learning PP methods in improving CTM forecasts is not fully established.

In the present study, we develop a machine learning-based PP system, designed for implementation in the Visual atmospheric Environment Utility System (VENUS; <https://venus.nies.go.jp/>) operational air quality forecasting system operated by the National Institute for Environmental Studies (NIES). The PP is constructed by training deep neural networks (DNNs) on quality-controlled and quality-assured observations from ground-based air quality monitoring networks and is intended to increase the predictive skill of the VENUS forecasts.

2 Materials and methods

Figure 1 presents an overview of the PP system developed in this study. For comparison, the schematic flow of forecast correction by data assimilation and emission inversion is also shown. In data assimilation and inversion, observational data are ingested to update the model initial conditions and emission fields sequentially, thereby improving subsequent forecasts. In contrast, PP is constructed in advance from historical pairs of forecasts and observations and acts directly on the model output to correct the forecast values. Because data assimilation operates upstream of the CTM, uncertainties from the representation of chemical and physical processes and from insufficient model resolution cannot be corrected until the next assimilation cycle and thus appear as errors or biases in the forecasts. In contrast, PP is performed downstream of the model and, if properly trained and designed, can compensate for these model-induced uncertainties when adjusting the forecasts.

2.1 VENUS operational air quality forecasting system

VENUS is an operational air quality forecasting system for the Japanese Archipelago that provides 7-day forecasts of surface ozone and $\text{PM}_{2.5}$ concentrations in the form of horizontal maps and regionally aggregated time series (Sugata et al. 2011). VENUS uses a triple-nested configuration comprising an East Asia domain at 45 km resolution (Domain 1; 137×87 grids), a broader Japan domain at 15-km resolution (Domain 2; 223×202 grids), and a Japan-focused domain at 5-km resolution (Domain 3; 352×346 grids). Operational products are distributed for Domain 3.

VENUS is built on an offline, one-way coupling between Weather Research and Forecasting (WRF) model ver. 4.1.1 (Skamarock et al. 2019) and CMAQ version 5.3.1 (U.S. Environmental Protection Agency 2019). Anthropogenic emissions were based on the Regional Emission inventory in ASia (REAS) ver. 3 (Kurokawa and Ohara 2020), supplemented by Hemispheric Transport of Air Pollution (HTAP) ver. 2.2 for sectors not represented in REAS (notably aviation and international shipping) (Janssens-Maenhout et al. 2015). Biogenic emissions were calculated using Model of Emissions of Gases and Aerosols from Nature 2.1 (MEGAN2.1) (Guenther et al. 2012), biomass-burning emissions were taken from Global Fire Emissions Database 4.1 (GFED4.1) (van der Werf et al. 2017), and volcanic emissions for Japan were prescribed using observational information compiled by the Japan Meteorological Agency (Japan Meteorological Agency, n.d.). Gas-phase chemistry and aerosol processes were represented using the SAPRC-07 mechanism (SAPRC07tc) and the CMAQ AERO7 aerosol module, respectively (Carter 2010; Byun and Schere 2006). For Domain 1, chemical boundary conditions were derived from Model for Ozone and Related Chemical Tracers 4 (MOZART-4) (Emmons et al. 2010). Further details of the model setup and sensitivity choices are provided in Itahashi et al. (2026).

2.2 AEROS

Hourly surface concentrations of ozone and $\text{PM}_{2.5}$ archived in the Atmospheric Environmental Regional Observation System (AEROS) (Ministry of the Environment, n.d.) were used as the observational target data for model training. AEROS is Japan's nationwide ground-based air-quality monitoring network and provides hourly measurements of six criteria pollutants: carbon monoxide, sulfur dioxide, nitrogen dioxide, photochemical oxidants (dominated by ozone), suspended particulate matter, and $\text{PM}_{2.5}$. As of 2023, photochemical oxidants (ozone) and $\text{PM}_{2.5}$ observations were available from 1,168 and 1,100 monitoring sites, respectively. To construct

gridded training datasets, we applied a two-stage quality control (QC) procedure and then mapped the validated observations onto a 0.375° latitude–longitude grid.

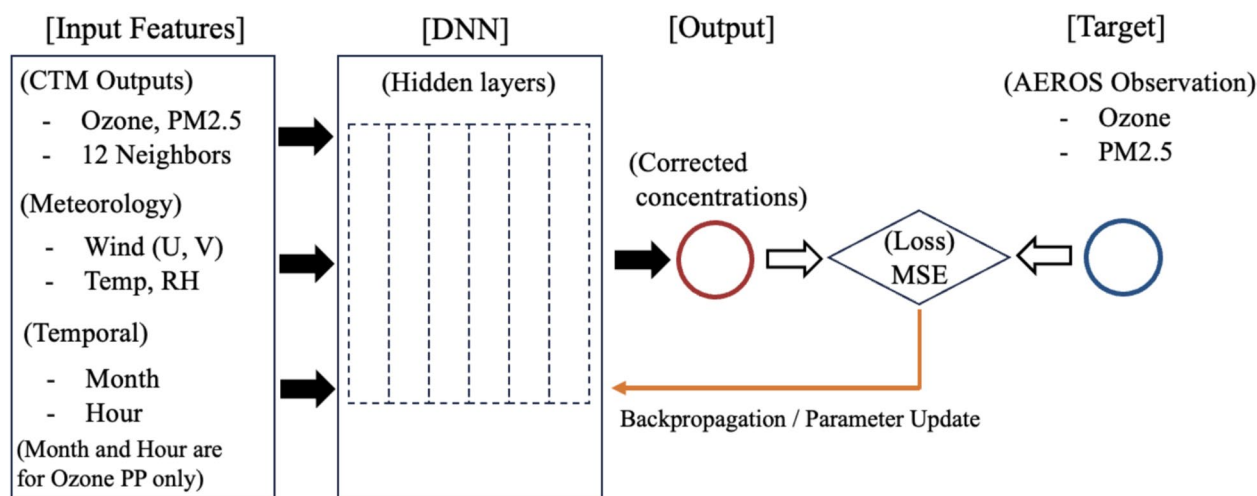
The QC procedure consists of two sequential steps, both of which were applied only when at least five neighboring sites were available ($M \geq 5$). First, a site-level screening was conducted to identify monitoring sites with limited spatial representativeness. For each monitoring site i ($i = 1, 2, \dots, N$), the monthly mean diurnal variation was calculated during 2015–2017 and compared with those of neighboring sites located within a 50-km radius. Let M denote the number of neighboring sites within this radius. For each site pair, the root mean square difference (RMSD) and the Pearson correlation coefficient (R) were computed using the monthly mean diurnal profiles, and threshold values for RMSD and R were defined as functions of inter-site distance. A site pair was classified as dissimilar when the RMSD exceeded the 90th percentile of the distance-dependent distribution or when R was below the 10th percentile. A target site i was considered to satisfy this screening when fewer than half of the M neighboring sites were classified as dissimilar.

Second, a time-level screening was applied to hourly observations to exclude temporally anomalous data. Outliers were identified relative to a climatological diurnal variation constructed from a three-year period (2015–2017). For each calendar day and hour, statistical values (quartiles) were derived using data within ± 10 days of the target calendar day (21 days per year, corresponding to 63 days over three years). An observation at time $t = T_0$ was defined as an outlier when it exceeded the upper inner fence (median + $1.5 \times$ interquartile range) derived for the corresponding calendar day and hour. A temporal window of ± 4 h centered at T_0 (i.e., 9 hourly values in total) was then considered, and the observation at T_0 was retained when fewer than half of the values within this window were flagged as outliers.

Only observations passing both QC stages were used to generate the gridded datasets. PP was developed using 144 grid cells for which the observational coverage was sufficient to perform the QC procedure. The grid resolution (0.375°) was selected to ensure that each grid cell typically contained on the order of five monitoring sites, providing adequate spatial redundancy for the QC procedure.

2.3 PP system based on machine learning

A DNN architecture was used to construct the PP system, and an overview is provided in Fig. 2. In this study, the training and test datasets were generated by retrospective WRF/CMAQ simulations configured consistently with VENUS, rather than by directly using archived



Independent model for each grid and species

Fig. 2 Schematic of DNN architecture for the PP system. Inputs include CTM-predicted concentrations of ozone and PM_{2.5}, meteorological variables, and temporal indicators (month and hour). These features are processed through hidden layers to produce corrected ozone and PM_{2.5} concentrations. During the training phase, ground-based observations from AEROS are used as teacher data (target). Individual DNN models are constructed independently for each grid cell and for each pollutant species

operational VENUS outputs. This was because the data required for PP development and evaluation could not be sufficiently obtained from the operational VENUS archive: the operational VENUS system has undergone configuration updates over time, making it difficult to prepare a sufficiently long and internally consistent dataset, and some variables required for PP development were not preserved in the archive. Thus, while this work is intended as a step toward implementation in the operational VENUS system, the present study serves as an evaluation of the PP approach within a VENUS-consistent framework under the current data constraints.

The predictor variables consisted of Domain 3 WRF/CMAQ outputs. To match the resolution of the observational target data, the WRF/CMAQ outputs from Domain 3 were regridded to a 0.375° latitude–longitude grid. The input features included model-predicted ozone and PM_{2.5} concentrations, zonal and meridional wind components, air temperature, and relative humidity. In addition, the month of the year and hour of the day were considered as input features. For ozone and PM_{2.5}, model-predicted concentrations in the 12 neighboring grid cells surrounding each target grid cell were also included as additional inputs to account for upwind transport and the surrounding horizontal concentration structure. During the training phase, ground-based observations from AEROS were used as teacher data (target), and individual DNN models were constructed independently for each grid cell and for each pollutant species.

In this study, the PP models were developed only for grid cells containing observation sites. As a first step, we prioritized establishing and validating the effectiveness of the PP approach at these grid cells, where direct evaluation was possible, rather than addressing spatial generalization over the full Domain 3. Extending the current approach to a spatially generalized correction framework applicable to the full Domain 3 remains an important subject for future work.

To investigate the effects of different input features on the PP system performance, we conducted sensitivity experiments focusing on specific grid points with a limited training period. Table 1 summarizes the results for PM_{2.5}. Although incorporating meteorological variables in addition to PM_{2.5} concentration improved accuracy, the planetary boundary layer height contributed little to this improvement. Using zonal and meridional wind data components contributed more to prediction accuracy than using wind speed and direction. Table 2 demonstrates the sensitivity of ozone prediction to temporal features (month and time). Including both variables increased accuracy substantially. This characteristic, which was absent in the PM_{2.5} results, implied that time and month were essential for capturing the diurnal and seasonal variability inherent in ozone concentrations. Consequently, the input feature set for the PM_{2.5} PP system comprised PM_{2.5}, ozone, 2-m temperature, zonal and meridional wind components, and relative humidity. For the ozone PP system, month and time were added to this feature set. The PP models were

Table 1 Summary of sensitivity experiments assessing the impact of meteorology-related input feature selection on PP performance for PM_{2.5} forecasting

Input features								Statistics	
PM _{2.5}	Ozone	RH	Temperature	PBL	Wind speed	Wind direction	U and V	Correlation	RMSD (µg/m ³)
Y	Y	-	-	-	-	-	-	0.68	7.23
Y	-	Y	-	-	-	-	-	0.70	7.16
Y	-	-	Y	-	-	-	-	0.67	7.30
Y	-	-	-	Y	-	-	-	0.67	7.35
Y	-	-	-	-	Y	Y	-	0.68	7.35
Y	-	-	-	-	-	-	Y	0.69	7.14
Y	Y	Y	Y	-	-	-	Y	0.73	6.95

"Y" defined as "applied". "-" defined as "not applied". RMSD: root mean square deviation defined in equation; RH: relative humidity; PBL: planetary boundary layer; U: zonal wind speed; V: meridional wind speed

Table 2 Summary of sensitivity experiments assessing the effect of temporal input feature selection on PP performance for ozone forecasting

Temporal input features		Statistics	
Hour	Month	Correlation	RMSD (ppbv)
-	-	0.88	8.38
Y	-	0.89	7.99
-	Y	0.88	8.15
Y	Y	0.90	7.59

"Y" defined as "applied". "-" defined as "not applied". RMSD: root mean square deviation defined in equation

implemented using the TensorFlow library (Abadi et al. 2016), and the network parameters were optimized using the Adam optimizer (Kigima and Ba 2017) by minimizing the mean squared error loss.

The PP system was developed using data from 2013 to 2016, and its performance was evaluated using 2017 data. In the validation phase, the PP system was used to correct the raw model forecasts, and its skill was quantified by comparing the pre- and post-corrected values against AEROS observations. In machine learning studies, model validation is often performed using random splits or random sampling, in which the full dataset is randomly partitioned into training and validation subsets. However, our goal is eventual implementation in an operational forecasting system, in which models trained on historical data are used to adjust current or future forecasts; therefore, random partitioning could be inappropriate. Random splits tend to mix temporally adjacent samples between training and validation, allowing seasonal and other time-dependent information to be shared implicitly, which can inflate apparent performance (data leakage). Therefore, we used a

chronological split for validation to separate the training and testing periods strictly.

In the present study, we did not separately evaluate the dependence of PP performance on forecast lead time. Because the available data did not allow a systematic lead time-dependent analysis, the dependence of PP performance on forecast lead time remains an important subject for future work.

2.4 Statistics

The performance of the PP system was evaluated using the following statistical metrics. We evaluated agreement using the Pearson correlation coefficient (R) and root mean square deviation (RMSD). Normalized mean bias (NMB) and mean fractional bias (MFB) are signed metrics that quantify the systematic tendency of the model to overestimate (NMB > 0; MFB > 0) or underestimate (NMB < 0; MFB < 0) observations. Normalized mean error (NME) and mean fractional error (MFE) are non-negative metrics that measure the overall magnitude of the deviations regardless of sign. The index of agreement (IOA) ranges from 0 to 1, with values closer to 1 indicating better agreement. The squared-error formulation means that the IOA is sensitive to large errors (e.g., high-pollution events). We also computed the Taylor skill score (TSS) (Taylor 2001), which combines R and the ratio of standard deviations between model predictions and observations and ranges from 0 to 1, with higher values indicating better agreement in pattern similarity (correlation) and variability (standard deviation). Following commonly used CTM performance criteria, we interpreted NMB and NME based on the ozone-related model performance evaluation proposed by Emery et al. (2017), and MFB and MFE with reference to the PM_{2.5}-related evaluation proposed by Boylan and Russell (2006).

$$R = \frac{\sum_1^N (O_i - \bar{O})(M_i - \bar{M})}{\sqrt{\sum_1^N (O_i - \bar{O})^2} \sqrt{\sum_1^N (M_i - \bar{M})^2}} \quad (1)$$

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_1^N (M_i - O_i)^2} \quad (2)$$

$$\text{NMB} = \frac{\sum_1^N (M_i - O_i)}{\sum_1^N O_i} \times 100 \quad (3)$$

$$\text{MFB} = \frac{1}{N} \frac{\sum_1^N (M_i - O_i)}{\sum_1^N (M_i + O_i)} \times 100 \quad (4)$$

$$\text{NME} = \frac{\sum_1^N |M_i - O_i|}{\sum_1^N O_i} \times 100 \quad (5)$$

$$\text{MFE} = \frac{1}{N} \frac{\sum_1^N |M_i - O_i|}{\sum_1^N (M_i + O_i)} \times 100 \quad (6)$$

$$\text{IOA} = 1 - \frac{\sum_1^N (M_i - O_i)^2}{\sum_1^N (|M_i - \bar{O}| + |O_i - \bar{O}|)^2} \quad (7)$$

$$\text{TSS} = \frac{2(1 + R)}{(\delta_m/\delta_o - \delta_o/\delta_m)^2} \quad (8)$$

Here, N is the total number of paired observation–model samples, O is the observations, and M is the model values (before and after PP). The corresponding sample means are $\bar{O}$ and $\bar{M}$ and the standard deviations are δ_o and δ_m , respectively.

3 Results and discussion

3.1 Ozone

Figure 3 shows R and RMSD before and after applying the PP and the corresponding changes for ozone for each grid cell. Grid cells shown in gray indicate areas where AEROS observations were insufficient to construct the PP. Before PP, R generally ranged from 0.5 to 0.7, and only 2 of 144 grid cells (1.4%) exhibited $R > 0.8$. After PP, R increased in all grid cells and exceeded 0.6 everywhere; the number of grid cells with $R > 0.8$ increased to 136 of 144 (94.4%). RMSD tended to be larger over mountainous regions than over urban areas. Before PP, 55.6% of grid cells (80 of 144) had RMSD values exceeding 20 ppbv, and all grid cells exceeded 10 ppbv. After PP, RMSD decreased substantially across the domain. No grid cell had RMSD greater than 20 ppbv, and only 5 of 144 grid

cells (3.5%) remained above 10 ppbv. Overall, R and RMSD improved in every grid cell after PP was applied.

Figure 4 shows the frequency distributions of forecast deviations (model minus observation) aggregated over all grid cells for ozone. Before applying PP, the distribution was shifted toward positive values and was positively skewed, indicating a systematic positive bias (overestimation) in the raw air quality forecasts. The mean deviation was 16.5 ppbv. Similar positive biases in CTM-based ozone predictions over East Asia, including Japan, have been reported in previous model intercomparison studies. For example, the Japan Study for Reference Air Quality Modeling (J-STREAM) (Chatani et al. 2020) evaluated summertime ozone performance over Japan using 33 model configurations and reported ozone overestimation across all configurations, particularly over urban areas. Similarly, the Model Inter-Comparison Study for Asia Phase III (MICS-Asia III) (Li et al. 2019), which compared model outputs submitted by 14 independent groups from China, Japan, South Korea, the United States, and other regions, reported that many models exhibit pronounced ozone overestimation along the western Pacific coastal regions. Only 32.7% of the samples fell within ± 10 ppbv of the observations, and 65.6% fell within ± 20 ppbv. After PP, the positive shift in the raw forecasts was largely removed, and the systematic bias was substantially reduced; the mean deviation decreased to -0.6 ppbv. The distribution also became more symmetric and narrower, indicating a smaller standard deviation. In total, 82.5% of the samples fell within ± 10 ppbv and 98.1% within ± 20 ppbv, demonstrating that PP achieved a marked improvement in forecast accuracy. Table 3 summarizes the statistics computed using all grid cells and all hourly samples. Applying PP increased R substantially, from 0.66 to 0.87. RMSD was also markedly reduced to below 10 ppbv. The IOA and TSS increased considerably, both exceeding 0.9. Bias-related metrics showed large improvements: the NMB and MFB, which were close to 50% in the raw forecasts, decreased to around 2% after PP.

Ozone is known to exhibit a pronounced diurnal cycle, with lower concentrations at night and higher concentrations during daytime. Therefore, evaluating PP performance separately for daytime and nighttime conditions is important for assessing whether the correction works consistently across different parts of the diurnal cycle. To examine this point, we calculated the statistics separately for daytime (06–18 JST) and nighttime (18–06 JST). Before applying PP, the mean bias was 15.5 ppbv during daytime and 17.5 ppbv during nighttime, indicating that the raw CMAQ forecasts showed a somewhat larger positive bias at night. After PP, however, the mean bias was reduced to -0.6 ppbv for both daytime and nighttime, indicating that the residual bias became very small and

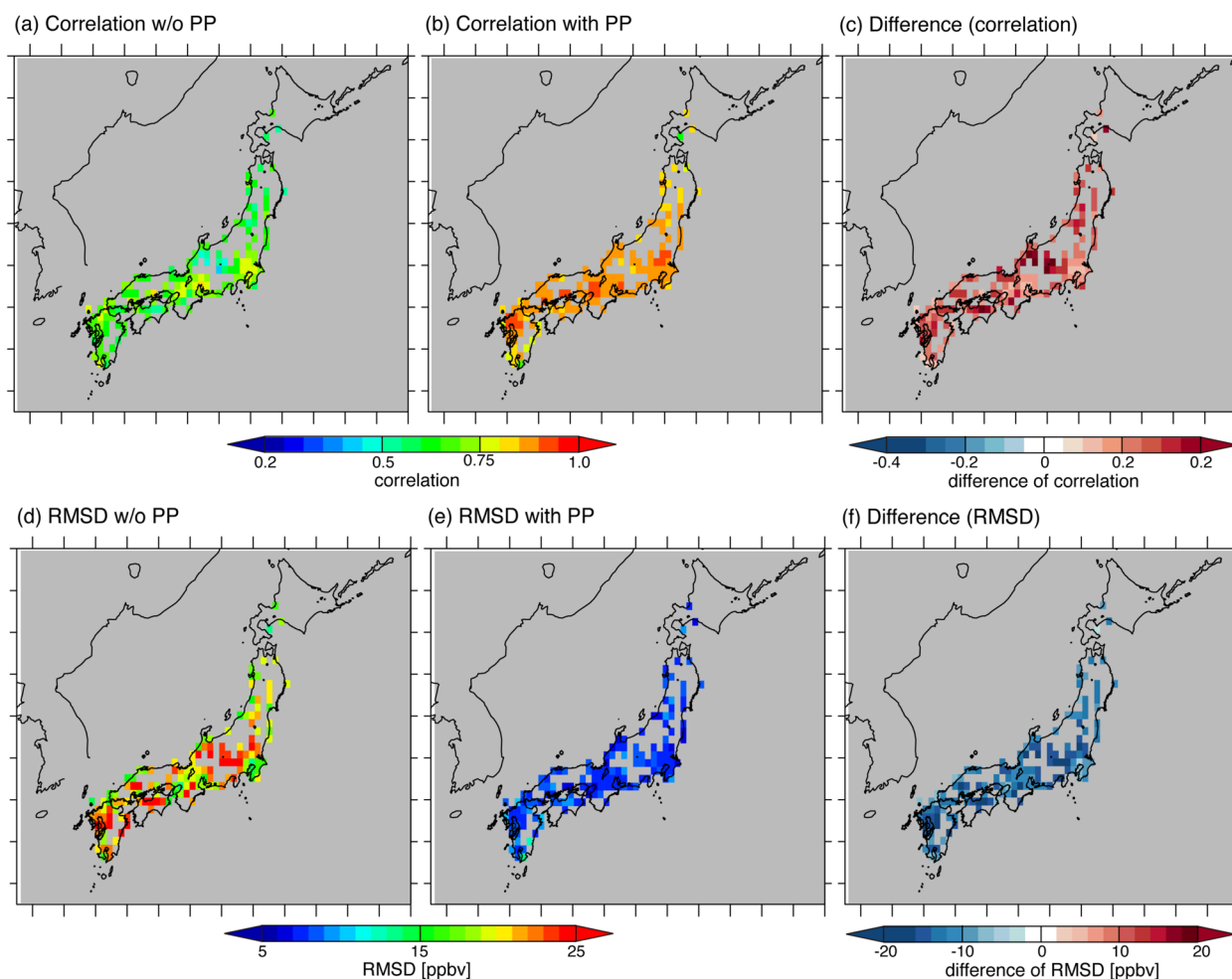


Fig. 3 Spatial distributions of R and RMSD for ozone before and after PP. **a** R before PP, **b** R after PP, **c** change in R (after minus before), **d** RMSD before PP, **e** RMSD after PP, and **f** change in RMSD (after minus before). Grid cells shown in gray indicate areas where AEROS observations were insufficient to construct the PP

nearly identical between the two periods. The correlation coefficient also improved substantially, from 0.71 to 0.89 during daytime and from 0.57 to 0.84 during nighttime. Similarly, RMSD decreased from 19.6 to 7.8 ppbv during daytime and from 21.7 to 8.4 ppbv during nighttime. These results indicate that the PP effectively reduced systematic bias and improved forecast accuracy during both daytime and nighttime, rather than improving performance only under a specific part of the diurnal cycle.

Based on a review of CTM studies, Emery et al. (2017) proposed performance benchmarks for ozone using NMB, NME, and R . They defined a performance goal of NME of $<15\%$, NMB within $\pm 5\%$ and R of >0.75 , and a more permissive performance criterion of NME of $<25\%$, NMB within $\pm 15\%$, and R of >0.50 . Figure 5 shows the grid-wise distributions of these statistics. The performance goal and performance criterion are indicated by thick solid and dash-dotted lines, respectively. A grid

cell was considered to meet a benchmark if its point lay within the corresponding envelope and the associated R threshold was satisfied. Before applying PP, all grid cells were on the right-hand side of the diagram, reflecting a pervasive positive bias. NME and NMB exhibited an approximately proportional relationship, a pattern commonly observed when forecasts are systematically overestimated. Under the raw forecasts, none of the grid cells satisfied the performance criterion. After applying PP, the distribution collapsed toward the lower-central region of the diagram, indicating substantial improvements in NME and NMB. Overall, 91.7% (132 of the 144) of the grid cells satisfied the performance criterion, and 2.8% (4 of the 144) satisfied the performance goal.

3.2 PM_{2.5}

Figure 6 shows the grid-wise R and RMSD before and after applying PP and the corresponding changes for

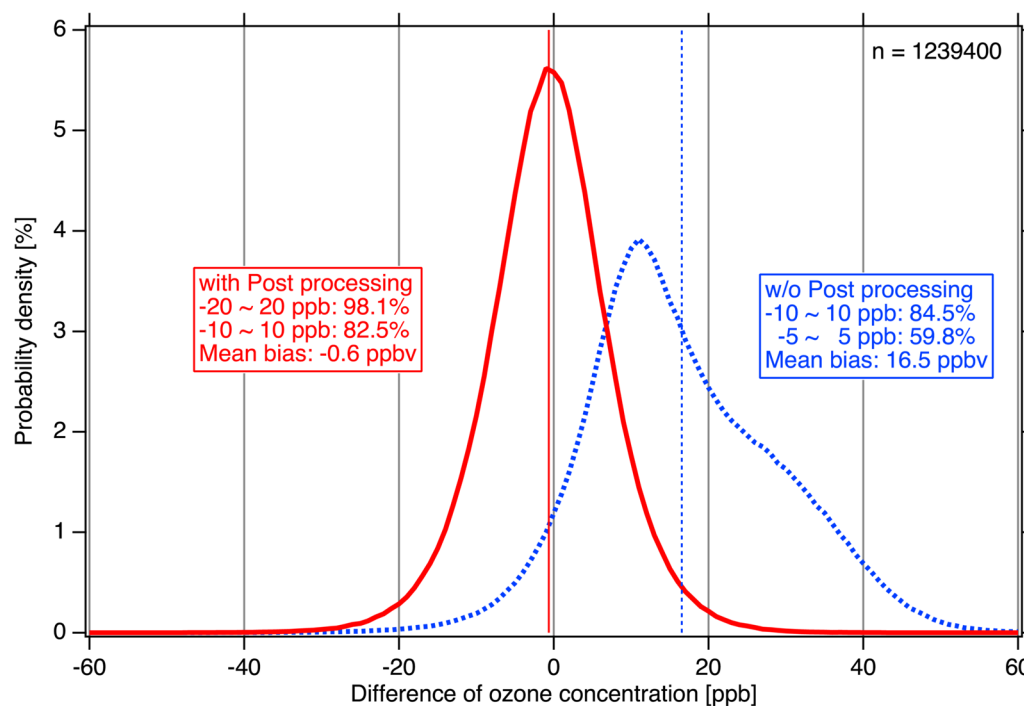


Fig. 4 Frequency distribution of forecast deviations (modeled minus observed ozone concentrations). The blue curve denotes the raw forecasts (before PP), and the red curve denotes the post-processed forecasts (after PP). The vertical blue and red lines indicate the corresponding mean biases. The fractions of samples with deviations within ± 20 and ± 10 ppbv are also indicated

Table 3 Summary of statistical metrics for ozone forecasts

$n = 1,239,400$	Mean bias (ppbv)	R	RMSD (ppbv)	NMB (%)	NME (%)	MFB (%)	MFE (%)	IOA	TSS
No PP	16.5	0.66	20.7	51.3	53.7	49.2	50.9	0.41	0.78
PP	-0.6	0.87	8.1	-2	19.2	1.9	24.7	0.93	0.92

$PM_{2.5}$. Grid cells shown in gray indicate areas where AEROS observations were insufficient to develop the PP. The mean R across the 144 evaluated grid cells was 0.55 before PP. After PP, R increased in every grid cell and the domain mean increased to 0.67. The number of grid cells with $R > 0.6$ increased from 45 of 144 grid cells (31.3%) to 124 of 144 grid cells (86.1%). Only one grid cell had $R > 0.7$ before PP, whereas 67 of 144 grid cells (46.5%) exceeded this threshold after PP. RMSD improved in all grid cells. The domain mean RMSD decreased from 8.8 to 6.4 $\mu g/m^3$. Before PP, RMSD exceeded 5 $\mu g/m^3$ in all grid cells; after PP, the number of grid cells exceeding 5 $\mu g/m^3$ decreased to 126 of 144 grid cells (87.5%). The number of grid cells with RMSD greater than 10 $\mu g/m^3$ also decreased from 10 to 6.

Figure 6d and e shows some grid cells with persistently large RMSD (e.g., in central to northern Japan and over the southwestern islands). Because these high-RMSD

cases were isolated grid cells without similarly large errors in neighboring cells, the corresponding R values were also locally low, and the behavior was not observed for ozone, we inferred that unresolved local aerosol emissions were a primary cause. Plausible contributors include volcanic emissions and agricultural open burning (crop residue burning). In addition, Fig. 6f shows grid cells where the RMSD reduction remained small even after PP. For these grid cells, there was a large discrepancy in the mean observed $PM_{2.5}$ concentrations between the training and validation periods, with the validation-period mean being approximately 40% lower than that during training. This inter-period shift in the observational baseline probably limited the effectiveness of the learned correction and was a primary cause of the small RMSD improvement.

Figure 7 shows the frequency distribution of forecast deviations (model minus observation) for $PM_{2.5}$

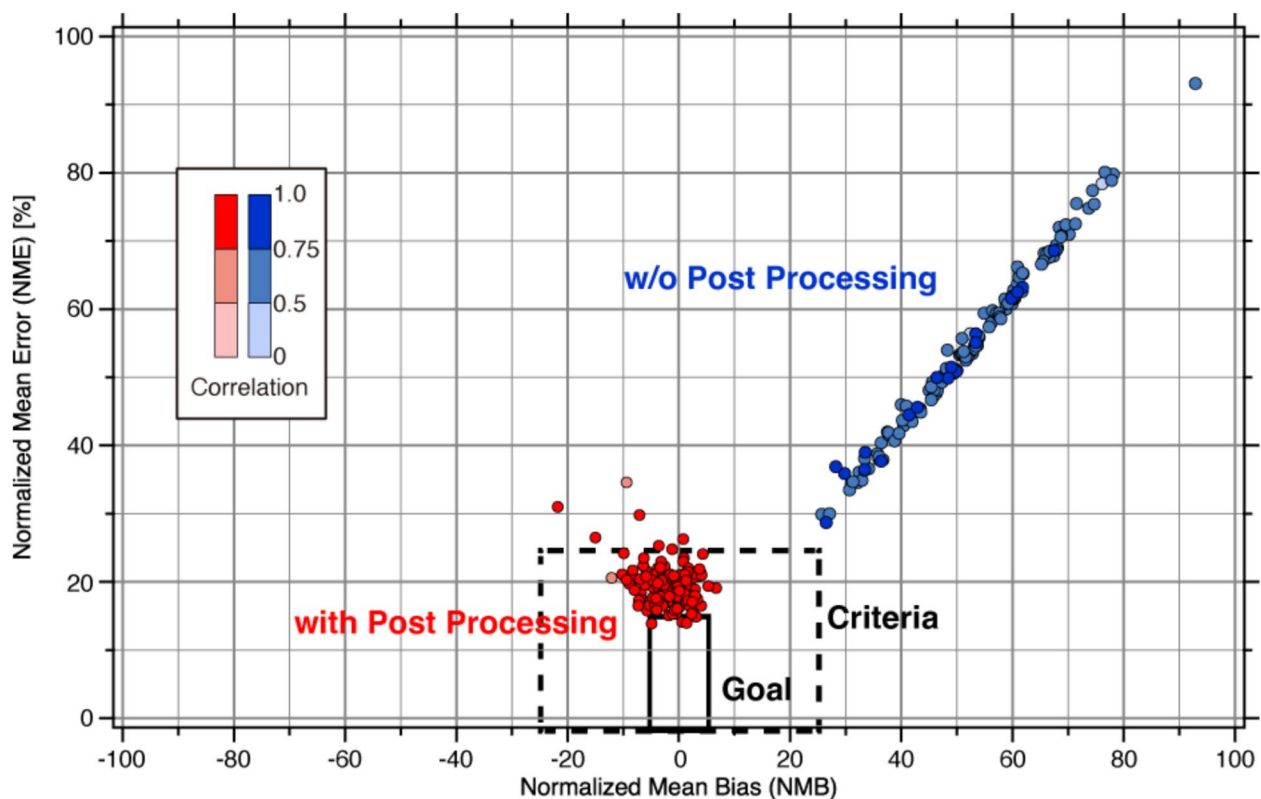


Fig. 5 Soccer goal plot of NMB versus NME for ozone. Blue markers indicate results before PP, and red markers indicate results after PP. R is shown by the color contours. The performance criterion (acceptable range) and performance goal (target range) proposed in Emery et al. (2017) are indicated by dash-dotted and solid boxes, respectively

aggregated over all grid cells. Before PP, the distribution was shifted toward negative values, indicating that the raw forecasts underestimated $\text{PM}_{2.5}$ concentrations, in contrast to ozone. The mean deviation was $-2.8 \mu\text{g}/\text{m}^3$. The fractions of samples falling within ± 10 and $\pm 5 \mu\text{g}/\text{m}^3$ were 84.5% and 59.8%, respectively. After PP, the distribution became narrower (smaller variance) and shifted overall toward positive values, consistent with a correction of the negative bias. The fractions within ± 10 and $\pm 5 \mu\text{g}/\text{m}^3$ increased to 92.9% and 68.8%, respectively, demonstrating improved forecast accuracy. Table 4 summarizes the statistics computed using all grid cells and all hourly samples. R increased from 0.54 to 0.67, whereas RMSD decreased from 9.0 to $6.5 \mu\text{g}/\text{m}^3$. Other metrics also increased, including IOA (from 0.69 to 0.79) and TSS (from 0.76 to 0.80), indicating that PP improved $\text{PM}_{2.5}$ forecast skill. However, the magnitude of improvement was smaller than that for ozone. The bias in the raw forecasts was underestimated, with a mean deviation of $-2.8 \mu\text{g}/\text{m}^3$, NMB of -24.7% , and MFB of -36.9% , consistent with the tendency reported for Japan in the 2013–2014 model intercomparison (J-STREAM) (Yamaji et al. 2020). PP compensated for this underestimation

(Fig. 7); nevertheless, the corrected forecasts had a mean deviation of $1.8 \mu\text{g}/\text{m}^3$, NMB of 15.8%, and MFB of 27.9%, indicating that the bias magnitude was reduced but the sign flipped to overestimation.

A likely explanation for this overcorrection was the decrease in $\text{PM}_{2.5}$ concentrations in Japan between the training and validation periods. Based on an analysis of AEROS data, Ito et al. (2021) reported that $\text{PM}_{2.5}$ concentrations declined by approximately 25% from 2012 to 2018. Consistent with this long-term trend, the domain mean observed $\text{PM}_{2.5}$ concentration decreased from $13.7 \mu\text{g}/\text{m}^3$ during the training period (2013–2016) to $11.2 \mu\text{g}/\text{m}^3$ in the validation year (2017), corresponding to a reduction of about 17.3%. Thus, although concentrations in Japan exhibited a clear decreasing trend, the PP developed in this study was trained statically using data from the training period and did not explicitly learn or correct for this year-to-year change (i.e., the long-term trend). Therefore, a PP model trained on earlier years with higher $\text{PM}_{2.5}$ levels probably overcorrected the 2017 forecasts, leading to an overestimation. The difference in mean observed $\text{PM}_{2.5}$ between the training and validation periods was $2.3 \mu\text{g}/\text{m}^3$, which was comparable to

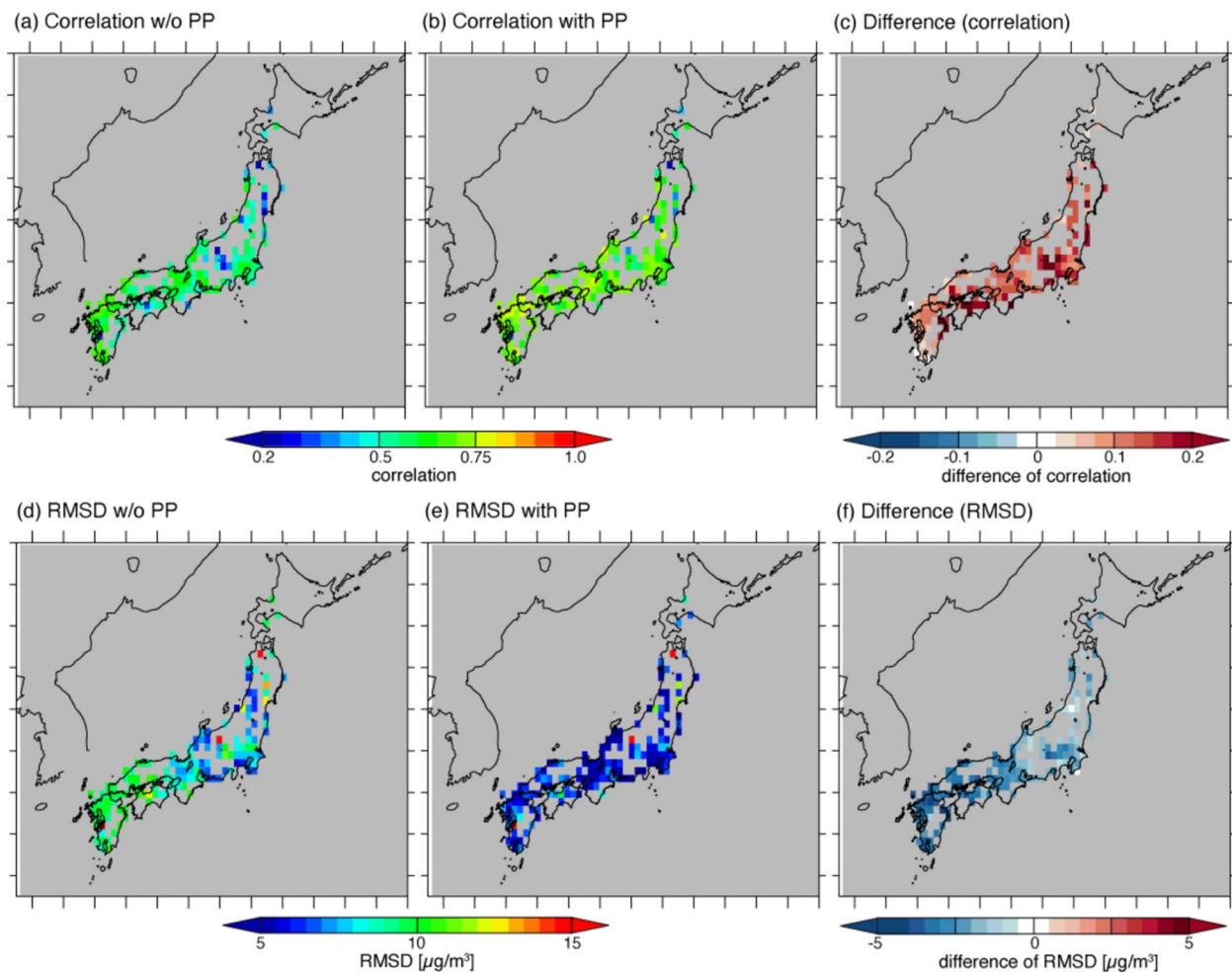


Fig. 6 Spatial distributions of R and RMSD for $PM_{2.5}$ before and after PP. **a** R before PP, **b** R after PP, **c** change in R (after minus before), **d** RMSD before PP, **e** RMSD after PP, and **f** change in RMSD (after minus before). Grid cells shown in gray indicate areas where AEROS observations were insufficient to construct the PP

the post-PP positive mean bias ($+1.8 \mu\text{g}/\text{m}^3$). This consistency supports the interpretation that the interannual decline in $PM_{2.5}$ contributed to the overcorrection. In contrast, Ito et al. (2021) reported that the observed ozone concentrations in Japan do not exhibit a decreasing trend. Consistent with this report, the mean observed ozone concentrations during the training period (2013–2016) and the validation year (2017) were 30.3 and 32.2 ppbv, respectively, corresponding to a modest difference of 1.9 ppbv (6.2%). This suggests that, compared with ozone, the smaller improvement achieved by PP for $PM_{2.5}$ forecasts were caused, at least in part, by the pronounced downward trend in $PM_{2.5}$ during the study period.

Boylan and Russell (2008) proposed performance benchmarks for $PM_{2.5}$ predictions by CTMs that are suitable for regulatory applications, defining the target performance goal and acceptable performance criterion ranges based on MFE and MFB. The performance

goal is MFE of $<50\%$ and MFB within $\pm 30\%$, whereas the performance criterion is MFE of $<75\%$ and MFB within $\pm 60\%$. Figure 8 shows the grid-wise distributions of these statistics. The performance goal and performance criterion are indicated by thick solid and dash-dotted boxes, respectively. A grid cell was considered to meet a benchmark if its point lay within the corresponding envelope. Before PP (blue symbols), most grid cells exhibited negative MFB, indicating systematic underestimation of $PM_{2.5}$. No grid cell achieved MFE of $<50\%$. Of the 144 grid cells, 66 of the 144 (45.8%) satisfied the performance criterion, whereas none satisfied the performance goal. After PP, the forecasts become overall positively biased, consistent with the discussion above. Nevertheless, MFE decreased substantially across the domain: 95.3% of grid cells (138 of the 144) satisfied the performance criterion, and 54.9% (79 of the 144) satisfied the performance goal. These

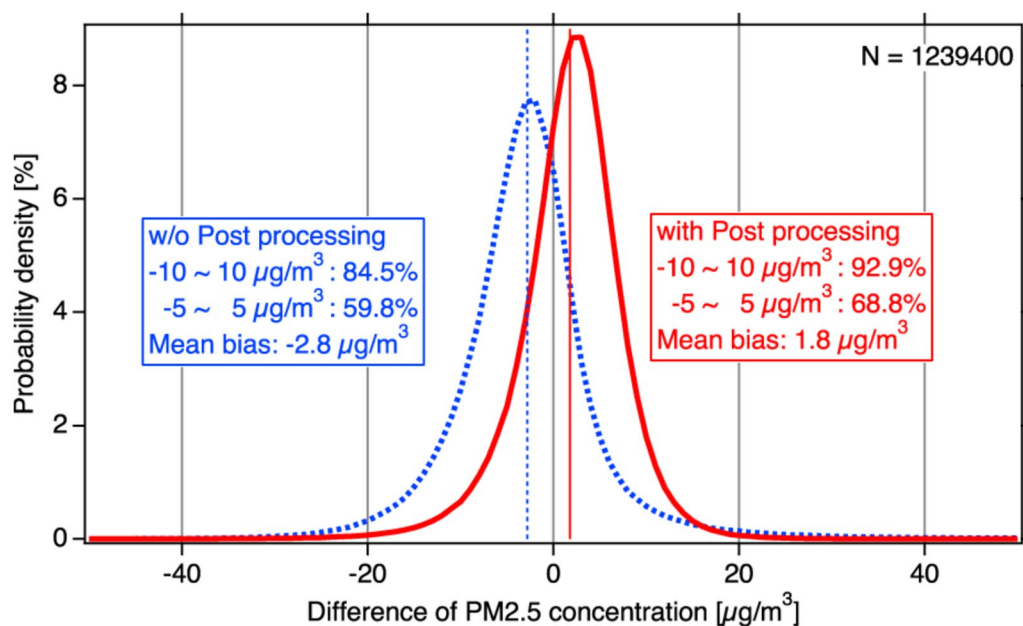


Fig. 7 Frequency distribution of forecast deviations (modeled minus observed ozone concentrations). The blue curve denotes the raw forecasts (before PP), and the red curve denotes the post-processed forecasts (after PP). The vertical blue and red lines indicate the corresponding mean biases. The fractions of samples with deviations within ± 10 and $\pm 5 \mu\text{g}/\text{m}^3$ are also indicated

Table 4 Summary of statistical metrics for $\text{PM}_{2.5}$ forecasts

$n = 1,239,400$	Mean bias ($\mu\text{g}/\text{m}^3$)	R	RMSD ($\mu\text{g}/\text{m}^3$)	NMB (%)	NME (%)	MFB (%)	MFE (%)	IOA	TSS
No PP	-2.8	0.54	9.0	-24.7	53.3	-36.9	72.8	0.69	0.76
With PP	1.8	0.67	6.5	15.8	40.4	27.9	46.9	0.79	0.80

results confirmed that PP improved $\text{PM}_{2.5}$ forecast skill substantially.

4 Conclusions

We developed a machine learning-based PP system for ozone and $\text{PM}_{2.5}$ forecasts within a VENUS-consistent modeling framework, with the aim of improving forecast performance for the current VENUS operational configuration. The PP system was constructed using a DNN trained on paired WRF/CMAQ outputs and ground-based observations (training: 2013–2016; evaluation: 2017). This study should therefore be regarded as an evaluation of the PP approach for the current VENUS framework, rather than as a directly deployable update to the operational VENUS pipeline.

For ozone, PP effectively removed the pervasive positive bias in the raw forecasts. The mean deviation (model minus observation) decreased from 16.5 to -0.6 ppbv, and the fractions of samples within ± 10 and ± 20 ppbv increased from 32.7 to 82.5% and from 65.6 to 98.1%,

respectively. Domain-wide skill also improved markedly (R : $0.66 \rightarrow 0.87$; RMSD: $20.7 \rightarrow 8.1$ ppbv; IOA and TSS > 0.9). According to the Emery et al. (2017) NMB and NME benchmarks, no grid cell met the acceptable range before PP, whereas after PP, 91.7% (132/144) satisfied the performance criterion and 2.8% (4/144) satisfied the performance goal. For $\text{PM}_{2.5}$, PP also improved predictive skill, increasing the domain-wide correlation (R : $0.54 \rightarrow 0.67$) and reducing RMSD ($9.0 \rightarrow 6.5 \mu\text{g}/\text{m}^3$). However, the mean bias changed sign after PP ($-2.8 \rightarrow +1.8 \mu\text{g}/\text{m}^3$), suggesting that the long-term decreasing trend in observed concentrations (about 17% lower in 2017 than during training) can cause overcorrection in a fixed PP model. Nevertheless, the PP-corrected forecasts met the benchmarks proposed by Boylan and Russell (2006) in most grid cells (performance criterion: 95.3%; performance goal: 54.9%).

The PP system developed in this study acts directly on model forecasts and thus can correct errors arising from uncertainties in physical and chemical parameterizations

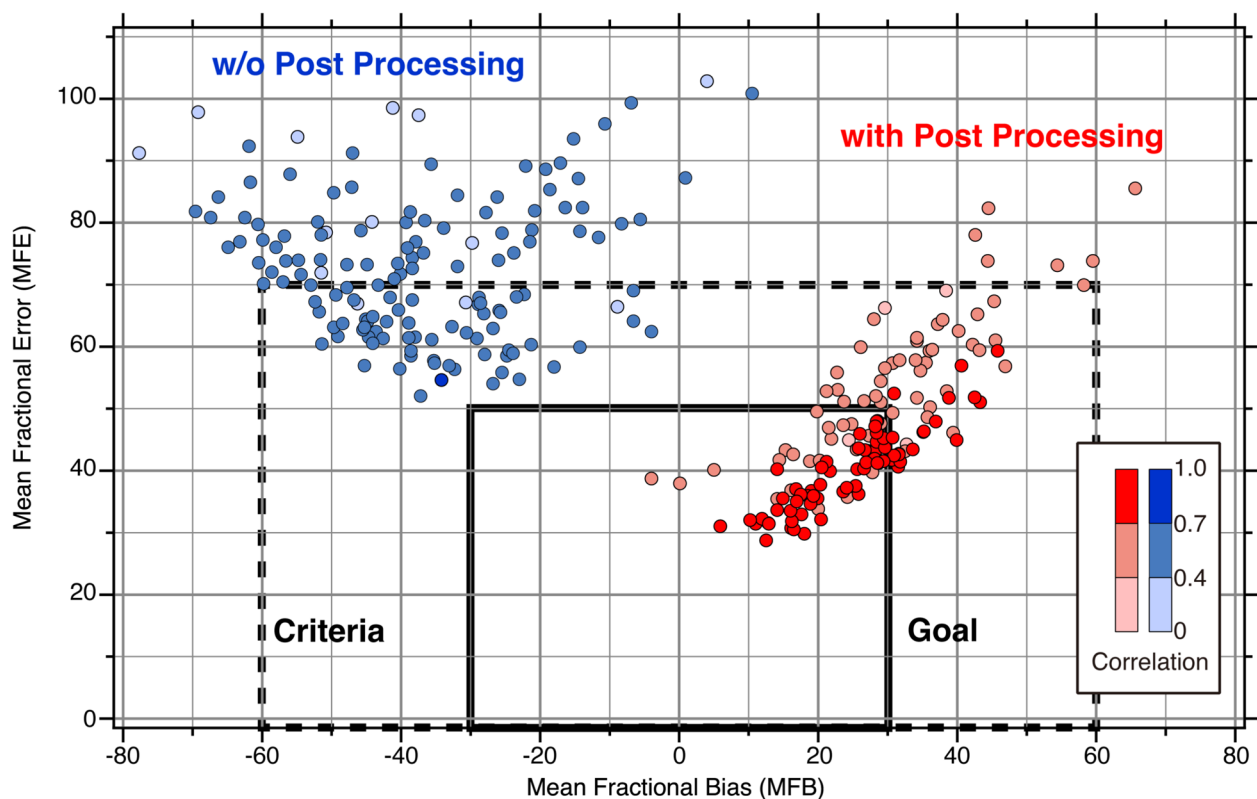


Fig. 8 Soccer goal plot of MFB versus MFE for $PM_{2.5}$. Blue markers indicate results before PP, and red markers indicate results after PP. R is shown by the color contours. The performance criterion (acceptable range) and performance goal (target range) proposed in Boylan and Russel (2006) are indicated by dash-dotted and solid boxes, respectively

and from limited model resolution. In addition, the PP system can be implemented and operated at substantially lower cost than upstream approaches, such as chemical data assimilation. These results demonstrate that PP is a practical and effective way to improve forecast skill for the current VENUS framework.

Although the effectiveness of the PP approach was clearly demonstrated, several challenges remain before it can be fully applied to an operational forecasting system. First, the present analysis was limited to 144 grid cells containing observation sites with sufficient observational coverage to support the QC procedure. For practical application, a spatially generalized correction framework over the full Domain 3 will be necessary. Extending the present approach to a spatially generalized correction framework applicable to the entire Domain 3 is an important subject for future work.

Second, the present $PM_{2.5}$ results indicate that a fixed PP model can become vulnerable to long-term changes in background concentration levels. For long-term operational application, adaptive strategies such as periodic retraining or sliding-window updating should therefore be considered. In addition, if the underlying VENUS

framework is updated in the future, the PP model should also be retrained and re-evaluated using outputs from the updated system.

Third, the dependence of PP performance on forecast lead time was not evaluated in the present study. Because forecast errors and the effectiveness of PP may vary with lead time, this should be investigated in future work ideally using actual operational forecast data from VENUS.

Overall, the present study demonstrates that machine learning-based PP can substantially improve ozone and $PM_{2.5}$ forecast skill under the current VENUS framework. Further development toward spatial generalization, adaptive retraining, and lead time-dependent evaluation using operational data will be important for routine operational deployment.

Abbreviations

AEROS	Atmospheric environmental regional observation system
CMAQ	Community multiscale air quality
CTM	Chemical transport model
DNN	Deep neural network
GFED	Global fire emissions database
HTAP	Hemispheric transport of air pollution
IOA	Index of agreement
J-STREAM	Japan study for reference air quality modeling
MEGAN	Model of emissions of gases and aerosols from nature

MFB	Mean fractional bias
MFE	Mean fractional error
MICS-Asia III	Model inter-comparison study for Asia phase III
MOZART	Model for ozone and related chemical tracers
NIES	National institute for environmental studies
NMB	Normalized mean bias
NME	Normalized mean error
QC	Quality control
REAS	Regional emission inventory in Asia
RMSD	Root mean square deviation
SAPRC07tic	SAPRC-07 chemical mechanism (tic)
TSS	Taylor skill score
VENUS	Visual atmospheric environment utility system
WRF	Weather research and forecasting

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Author contributions

KY proposed the topic, conceived and designed the study, and prepared the original draft. SI and SS developed the methodology and software tools and performed data curation. MH contributed to the investigation and data analysis. SI, MH, and SS collaborated with the corresponding author in the review and editing of the manuscript. All authors read and approved the final manuscript.

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Data availability

Data will be made available upon reasonable request.

Declarations

Competing interests

The authors declare that they have no competing interest.

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